

ActInf Livestream #019.0 ~ Deeply Felt Affect: The Emergence of Valence i

Livestream | Deeply Felt Affect: The Emergence of Valence in Deep Active Inference | 1:29:36

hello

welcome to the active inference live

stream this is active inference live

stream

number 19.0 on april first two thousand

twenty one

welcome to the active inference lab we

are a participatory online lab

that is communicating learning and

practicing

applied active inference you can find us

at our

links here this is recorded in an

archived live stream so please provide

us with feedback so that we can improve

on our work

all backgrounds and perspectives are

welcome and we'll be following

good video etiquette for live streams as

outlined in this checklist

this is the schedule that we've had so

far for 2021

and the upcoming two discussions on

april 6th

and the 13th will be number 19.1.2 on

this

paper that we're going to be discussing

today so right now

in 19.0 the goal is to set the context

and give a little background for 19.1

and 19.2

and people who are just learning about

the paper which is deeply felt affect

the emergence of valence in deep active

inference

by hesp smith parr allen firsten and
ramstead in 2021
and this video is definitely just an
introduction to the ideas
it's meant to be a two-way on-off ramp
from active inference so exposing the
active inference community
to ideas outside of active inference and
people who might be interested in some
of the bigger ideas connecting them to
the active inference implementations
and we have a bunch of sections planned
for today in 19
so thanks a lot to stephen and to blue
for
helping with the slides and for coming
on
today so maybe we can just um
introduce ourselves briefly and then we
can go
on to the rest of the slides we have
prepared so
i'm daniel i'm a postdoc in california
and i'll pass this stephen
hello i'm stephen i'm in toronto i'm
doing a practice-based phd
around social topographies and
community-based development
and i'll pass it over to blue
hi i'm blue knight i am an independent
research consultant based out of new
mexico
cool well i think this was a fun paper
for us to
read and we can just go right into
what were some of the big points of the
paper and the background ideas
before we go through the figure and the

model

which will take up a lot of the time the

paper

is called deeply felt affect the

immersion surveillance and deep active

inference

and it was published in the journal

neural computation

in 2021 and so i'll read the first one

and then

either of you can give a thought on that

in this letter meeting paper we

demonstrate that hierarchical

deep bayesian networks solved using

active inference

afford a principled formulation of

emotional valence

building on both the work mentioned

above as well as prior work on other

emotional phenomena

within the active inference framework

yeah so that's just given us that

insight into this idea of valence so

we'll be speaking about that more but

that's uh

that's that's an important part of this

paper

i'll say the next one our hypothesis

is that emotional valence can be

formalized

as a state of self that is inferred

on the basis of fluctuations in the

estimated

confidence or precision an agent has in

her generative model of the world that

informs

her decisions

cool so we're going to be connecting

some of the decision-making
aspects of agents and inference aspects
of agents
to emotional balance and they're going
to demonstrate it
so those are the goals and the claims
that they set out in the paper
any thoughts blue or do you want to go
to big questions and maybe introduce a
big question i'll introduce you to the
questions
all right go for it
so the big questions of this paper were
how can we make
more active inference models or agents
that are expressive
powerful realistic and interpretable
what are the implications of this active
inference
formalizes our survival and procreation
in terms of a single imperative
to minimize the divergence between
observed outcomes and phenotypically
expected
i.e preferred outcomes under a
generative model that is fine-tuned over
phylogeny and ontogeny
thoughts on this well the first one
this to me is about just developing the
active inference framework
to capture more of the patterns that we
see
in nature or in designed systems so how
do we make
systems that plan for multiple time
steps that's deep active inference
how do we make agents that interface
with the environment

so that's sort of a big move overall
and then this is going to be zooming in
on the valence
and some of those affective aspects
that's what
on the first point and then the second
this is a quote from the paper
and it says something that active
inference
does it formalizes our survival and
procreation not in terms of evolutionary
fitness again you could agree or
disagree with the authors or think it's
consistent with fitness or not
but claiming that the imperative is to
minimize
a specific kind of a divergence and so
what are the implications of that
if you think okay well if that's the
imperative then where's the situation
where we've been acting like there's a
different imperative
and is that consistent or inconsistent
with
um this and then the last quote and then
someone could give a dot
is a quote where they say we conclude
with a discussion of the implications of
this work
such as the relationship between
implicit metacognition and affect
connections to reinforcement learning
and future empirical directions
so what does active inference or this
model say about cognition or
metacognition
and reinforcement learning
yeah if i could add something i think

it's it really opens
up and a lot of the philosophical
questions and starts to give a modeling
interpretation so how the body comes in
how so the emotion starts to tie in to
a way that the body comes in an n-active
inference starts to become modelable
and it starts to also show how
affect is somehow connected
um to all cognition
and all processes so it's not some
weird artifact of shame
which might be more of a freudian trap
but it's actually an integration
of the cybernetic process all the way
through
in some way very nice it's also captured
a little bit on this slide which is
recycled from some
2020 vintage active which is
asking about how active inference
is going to be saying something about
the relationship between agents in the
world
and we're imagining an agent not a
one-step
tic-tac-toe player but we're thinking
about an agent that's imagining through
deep time
and they have imagination or the
capacity to simulate
and then they have affect and they can
also predict their affect and have
precision
and so on so here's sort of the drama
masks for the positive and negative
affect and then there's sort of
different world trajectories that are

being estimated through time
that lead to positive or negative
sentiment
and that gets around some of the sort of
basic considerations like in the paper
they say
well wouldn't the agent as they
meet their demise wouldn't they have
precision over their sensory states
and the answer is like yes but it's not
part of a long-term trajectory that
involves
desired preferences so by appealing to
model nesting
some of the sort of single layer
not paradoxes but just scenarios that
models might fall into like always going
with what was known before
never going what was known before these
can be resolved
just with empirical modeling interesting
one reason why it matters and we'll come
back to this
when we quote the authors later is that
these kinds of
richer models lead more closely to
experimentally tractable and unique
predictions of active inference
because if the only thing the model
predicts is the reaction time
that might fit the distribution better
than some other model now if it can also
fit the reaction time
and the eeg and also the self-report
it's starting to become a very rich
multi-modal
and multi-measurement way of talking
about behavior

which is what a good instrument should do

so this is helping us come closer through these contact points with other big important areas of research like reinforcement learning and autonomous computer agents and just inference agents of all kinds especially ones with action

but then maybe some of these other areas that we'll bring into the discussion because if we can have a scale-free framework that does apply to cognitive agents and then we frame it in terms of the input and output and the actions and the policy the generative model all of these terms maybe some of these other systems like online teams might be able to also be said to have at least computationally affective charge or precision or different things like that so we kind of talked a little bit about that in our 2020 paper for online teams any thoughts on sort of the bigger topics that this paper is up against before we jump into the abstract and the figures and everything

if i could just say one one thing i think that's very interesting is that it opens up um the the realities of trying to bring in the mind the body and the environment and there's been this question well how

do you do that and
why didn't we do that before was partly
because we wanted to be able to measure
things and lock things down
and and the question of how do you bring
this whole dynamic in how could you
possibly get a handle on all of that
um well emotion seems to actually make a
lot of sense in terms of
or let's say valence should say valence
more broadly
it makes sense in terms of how it could
be calibrating
the more conscious levels of that
process
so i think that has implications
for scaling psychology and bringing
consciousness back into
psychology and helping to
really move into a ecological
and ecosystem approach to psychology
lou any thoughts or do you want to start
with the first part of the abstract
i'll hold my thoughts i have somebody
but i don't want to ruin the surprise of
the paper so let's start off on abstract
agreed fun times all right um does
anybody want to start with the first
part of the abstract
i'll take it the positive negative
access
of emotional valence has long been
recognized as fundamental to adaptive
behavior
but its origin and underlying function
have largely eluded
formal theorizing and computational
modeling

using deep active inference a
hierarchical inference scheme that rests
on inverting a model of how sensory data
are generated
we developed a principal bayesian model
of emotional valence
this formulation asserts that agents in
further valence state
based on the expected precision of their
action model
an internal estimate of overall model
fitness
which the author's term subjective
fitness
this index of subjective fitness can be
estimated within any environment
and exploits the domain general
generality of sorry my screen is in the
way
second order beliefs which are beliefs
about beliefs there
nice so in this first part of the
abstract
kind of priming what this whole research
agenda
is about i'm seeing a few threads
one is about fitting good models
overall model fitness especially with a
subjective
twist so that's another kind of a bonus
that it's actually not just
objective model fitness it's subjective
so it's about
good fitting models with this
recognition that that's like
personalized
okay so that's cool thread then there's
the first words which are about the

positive and negative
aspect of emotional valence which is
despite the richness and the amount of
experiences and
all the diversity of altered states of
consciousness and adjectives
there's also kind of this um approach
versus recede binary
and um this is as they're saying it's
fundamental
and we see it in living forms like go
no go signaling and we also see it in
computational agents
and so this is putting a lot of
different kinds of systems
that have very nuanced maybe behavior
in their outputs but at a top level they
act
as if they have a sort of go no-go that
in a way
helps them be adaptive in their own
niche
and then the way that they're going to
go about that is actually with a second
level or second order
cybernetics model which is by enabling
belief about belief
maybe some of these gaps between systems
can be bridged
so interesting research questions that
are not always brought out because the
abstract is so reduced and they have to
say what they did do
but for those who are interested in
those questions that's good to know
all right i'll read this one and then
anyone can give a thought
we show how maintaining internal valence

representations

allow the ensuing effective agent to

optimize confidence in active selection

action selection preemptively valence

representations can in turn be optimized

by leveraging the bayes optimal updating

term for subjective fitness which we

label

affective charge or ac ac

tracks changes in fitness estimates and

lends a sign

to otherwise unsigned divergences

between predictions and outcomes

that's pretty interesting we simulate

the resulting affect of inference by

subjecting an in-silico

in the computer affective agent to a

teammate's paradigm requiring context

learning followed by context reversal

this formulation of affective inference

offers a principled account

of the link between affect mental action

and implicit metacognition it

characterizes

how a deep biological system can affer

its affective state

and reduce uncertainty about such

inferences through internal action

i.e top-down modulation of priors that

underwrite confidence

so this for me like was a big surprise

of the paper like the

the valence and um the relationship with

valence

and subjective fitness so or precision

accuracy

this was just a really different take

like compared to the last

paper that we did just like on valence
was very much like positive or negative
in terms of like feeling or emotion
where this is like really positive or
negative just based on
like surprise essentially like what if
my model doesn't match my expectations
like or or the outcomes if my
expectations don't match my outcomes
that is negative valence which that was
a kind of surprising take on this for me
compared to the last valence paper that
we did and the other thing
is really this the way that they've
modeled
this is totally like a perfect example
as they talk about in the discussion
of the bayesian brain and why bayesian
statistics
and updating um like especially when
they talk well
i'll hold i'll hold my thought for them
for that time but but really this was
just an exemplary paper for that for me
nice yeah one one thing that came up for
me there is it talks about
you know representations again and those
so
this idea that internal valence
representation so suddenly we've got a
way to bring
representations back into
um what had become quite a
non-representational
modeling approach and that's really
interesting because it's like normally
you think of representations previously
like the most concrete

is the thought then there's the body and
there was this fuzzy thing called
emotions which somehow was there and
axiology and all this stuff
but now it's like well wait you've got
an internal
valence representation which could
inform the body and the cybernetic
thinking metacognition process so it it
gives a way in for representations but
it's not representations as we
normally maybe approach that term so
that's quite
cool all right third abstract
who would like to read it
okay i'll read this one thus we
demonstrate
the potential of active inference to
provide a formal and
computationally tractable account of
affect
our demonstration of the face validity
and potential utility of this
formulation represents the first step
within a larger research program
next this model can be leveraged to test
the
hypothesized role of valence by
fitting the model to behavioral and
neuronal responses
nice so i set up the problem by being
about
using higher order cognition this second
level modeling
to approach some of the interesting
dynamics of
agents like their apparent
affective charge influencing their

behavioral decisions in the present
moment based upon
inference about the future and they're
going to demonstrate a simulation
there's going to be kind of a model
layer and then there's going to be a
simulation that goes down into the t
maze and a mouse and so there's sort of
two
levels of generality that we're going to
go through here
they start the paper with the
qualitative but still
rigorous level of generality which is
just talking about
action in different systems and
multi-scale biology
and the need to integrate bayesian
approaches
into how we think about evolution and
development
ecology and so on and then they
turn directly to the methods and perform
this
four-step walkthrough that will be
somewhat familiar to those who have seen
model stream one with ryan smith and
christopher white
where they also step through the
construction of this model but
it's something that as we'll probably
see soon it's it's just always worth it
to kind of build up this model
incrementally
then they with this implicit
metacognitive model
at step four of their models elaboration
they discuss a few kinds of

important key terms that are like
bridges between
previous computational psychological
models
and their active inference model
and then they implement a simulation
with a rat in a team ace which we'll
discuss as well
and then they bring up all these very
interesting questions
towards the discussion section of the
paper some of which we've exerted
and conclude with a discussion on the
future empirical directions
so before we go into the affect and
valence
model simulation implications any
thoughts on the roadmap
all right let's go to affect and valence
this is quoting from their
paper so definitely read it to see
everything that they say
but they refer to classical constructs
of valence
so maybe here just representing the
classical valence construct
on a kind of clock chart what would
either of you
think or say about valence or what
motivates the study of valence
this way or overall
so i just had a comment and it's based
on something that
uh the authors had said about
surveillance so here
uh we've got the arousal and like high
and low arousal on one axis
and positive and negative on the other

axis but the authors had mentioned that uh some people see valence as like a an array of positive like it's a positive and negative or two separate axes of valence and i thought that that was very interesting whereas like like it's a it's not a um you know a binary choice positive and negative but there's a range of positive and there's a range of negative and ambivalence like contains some degree of both i thought that that was an interesting uh take on valence i've thought about before cool and they bring up good scholarship with bringing up a uni-dimensional view so that would be like yes there's like a thermometer and it's kind of just like it goes up or down it's a number that has to be strictly ordered and less than greater than maybe even linearly correlated with a hormone or a chemical or a neural firing in some region or a pattern of activity then maybe there's multiple dimensions maybe there's um like as here an arousal and a high to low arousal and high to low valence positive negative um maybe it's multi-dimensional so what's cool about the active inference framework is we could test the computational roles of these kinds of variables and ask okay

this is
fitting the data better but is it um
more parameters is this worth the
trade-off that we're getting here
so we're kind of taking this single
dimensional model and thinking about it
with a family of other models
where we could test the influence of
other factors and see if
such patterns are detectable in the data
or ask what kind of
behavioral settings or groups or
questionnaires or whatever
it might be would elicit um a
differentiation on another
hypothesized axis yeah and just one
thing that goes with that is
it sort of shows you the current ways to
get at this
is kind of through self-reported
emotions
and using this kind of graph and
actually use this
a tool with our community with
disabilities for
um reporting back on their experiences
but you you're essentially still getting
this emotion
um as as a self-report and trying to
track back
um using sort of assumptions about how
that
plots with other self-reported emotions
and
um whereas the the and this is also used
a lot
in uh market research for marketing and
advertising this tool

so but actually in this
approach here they've got a way to
access this
without just getting self-reported words
um so this is this this i think
shows how it has you know this this is
an area that's got a lot of
application potentially cool
here is a slide just upon uh
differentiation
in english between affect
and effect so not to split hairs or to
introduce words that they didn't use in
the paper like effect
but what would be the point on this
stephen
well you've you've got just to give the
sense that it is often
confusing those two terms so if we have
the
the person these two people on the left
so you've got um
the person x has experiences and effect
some sort of you know relationship
to person wine it's it's created some
affect now that could be anger
if it was reported later and they said
what was their emotional state
and this affect contributes to a policy
selection
where person x interacts with person y
and pushes them into the pool so you can
see that
okay so that that um the effect
of this is that that person
y ends up in the pool so there's there's
a there's a there's an act and there's a
cause effect type of um linear

type of mechanical action that happens
um and of course
then there can be additional affects on
both the parties
and effect resulting from this
particularly if the person was to uh
you know drown or something depending on
the scenario so
i think it shows that there's a kind of
one's more dynamical and relational and
one is more
objective i suppose you could say and
cause and effect related
sounds good and here is from a previous
um discussion i think number
active 11 where we talked also about
affect
and about trajectories through affect
space so whether or not it's like how
things quote really are
it might be a behavioral index or
summary statistics of of someone's
online activity
even if you're not directly measuring
them or getting their self-report
and there can be trajectories through
this space
now one topic that um a sort of
detail that's a little bit special for
active inference that
doesn't get emphasized a ton is that
basically
the certainty in the model is
one and the same thing as having better
affect
better affective charge increasing
certainty and high certainty in the
model

are taking the place in a reward centric
of highest expected for example value
and it's one major difference between um
an active inference approach to thinking
about economics or decision making
versus a sort of just like classical um
maximum likelihood
reward maximization so in
these active inference frameworks
they're not modeling
a happy to sad variable and calling that
affective charge they're actually
modeling a precision variable
and then saying that when things are
more precise and are getting more
precise there is
low negative affect so it's kind of like
positive and then anxiety like negative
affect
arises as a function of
uncertainty in the model and
um here's just some words and some
thoughts that we had
put on anxiety from a previous one but
any thoughts on this about the sort of
integration of valence and certainty
rather than valence and expected
average what that means or how that's
different than other models that others
have seen
yeah well i think that does have quite
important implications
because it's now prospective it's kind
of what was the
contextual expectation on where things
would go
um as opposed to just it being this kind
of static

expectation on the moment itself
and it's all collapsed down into sort of
some sort of reductive
there's the idea that there's the
person's valence
is dynamic in relation to
um you know their expectation but then
there's also the precision at which that
expectation then
resolved or didn't resolve you know so
you may expect things to go badly
and then things went worse than you
expected or it went better than you
expected but it still might have gone
badly
but it didn't go as bad so it gives you
that
much more um well i don't know i
wouldn't use the word realistic i don't
know what term you'd use for that
no yeah that question cool blue
so i can't remember which paper it was
exactly but it was
talking about um anxiety it was one that
we did on the live stream talking about
anxiety as like a fixation
on the future right and i talked about
depression as a fixation on the past
was it the big five i feel like it was
the big five paper
um but but that is just an interesting
um
and kind of different perspective on
anxiety as as relative to the valence
that's presented here
cool and uh the question of how to map
model parameters to experiences that we
have which are

encultured and which are relating to our individual

kind of qualia that's a big jump too

so we shouldn't just instantly also

paper over the computational

psychiatric uh gap but that's what we're

going to unpack is what does the model really

say and then what is that going to help us do or rethink

one note is that they wrote that they are

referring to their model distincting it from just sort of active inference more broadly as

affective inference because it sort of has that

the active in there but it also has affect in there

and they're saying there's what makes it different

than just general active inference and

uh one of the one attribute that they

apply here is deep so that's meaning deep through time

or potentially also using deep uh

learning like deep neural network

approaches we had i think christopher white or someone

it all blurs together but somebody gave a few senses of deep

active inference and then

they're also modeling affect

and so that's why they call it affective inference so it's deep affective

inference

so if you know what active inference kind of is about

they're taking that skeletal framework
and extending it through time
and adding an affective
parameter or um a
set of interacting variables way to
think about
and model the kinds of things that
people study affect
in that's what they want to study with
this new development
so that's kind of what i think they're
doing in this paper
and um it's
also a little more general than just
affect because
as they say it can generalize to
potentially other higher level
explanations
for conditional dependencies between
higher level parameters and
how they reflect behavioral outcomes so
that's a pretty
general statement that's very important
though it's saying that
it could be possible to have other
higher hidden states
that cognitive ideas let's say and those
don't have to be just the positive or
negative
approach or recede that they model here
it could be something that's a lot more
subtle or
symbolic or categorical rather than this
is more of like
foraging go no go so this is a initial
foray into this kind of deep and hyper
parameter reaching back down
affective in that sense and it's tied to

the affect

rather than to the for example cognitive
work

but it's as they say implicitly

metacognitive and stephen as you said

it's not that affect is late in late

late and then like strikes into the
scene

it's actually that there's this sort of
fundamental cybernetic role

for affect regulation so it's not a red
flag

mode that is explosive or something it's
something that's

intrinsic like metabolism

okay they uh bring up a lot of previous
work and some limitations of the model
so check out the paper more if you want
to learn

about for example if you think well
that's a very narrow

uh model definition of valence they
might bring it up

in their paper so check that out but
let's just get right into

what the model is and

there's going to be sort of two acts
coming up here there's going to be

the model which is just the mathematics
of what they lay out and the formalisms
they lay out

and then we're going to go into the
simulation with the mouse or the rat
or the ant or whatever in the t mace so
the first is going to be general

and the second is going to go into a
simulation that they're going to get
results from as well

so we're going to walk through kind of
four clicks in the elaboration
incrementally as they write on an active
inference model

okay so fun stuff thanks everyone also
for

joining live so here we are in figure
one

who would like to go first maybe blue
you annotated this really well so what
did you see in figure one with model one
and model two

sure so model one is um
the generative model of perception which
we've seen before where s is
hidden state o stands for observations
 d is prior belief uh a

is this likelihood mapping or like a
probability distribution

and then τ is time which we've seen
before so

in m_1 there's no time this is the first
step of a generative model and it's just
perception so it's it's like sensory
input

and the posterior over hidden
states is all that's really taken into
the model of perception and then the
 m_2 so as the models as we go through
these m_1 m_2

m_3 and m_4 in the next few slides
there's an increasing layer of
complexity as we transition from m_1 to
 m_2

to m_3 and then m_4 with m_4 being the most
complex

and so in m_2 which is the generative
model of

anticipation the idea of perception
is then extended forward and backward
into the past if
you want to go there using a transition
matrix which is this
 b sub t for to represent the hidden
state
so it's just an extension of of
perception
essentially and it includes the
anticipation what you expect to happen
as opposed to the just the sensory input
which is just
what's present in m_1 so it's
anticipating hidden states over time
 m_2 nice
awesome good explanation
and we can look at some of their or
anything else to add actually or
ask here stephen on figure one was
really helpful i mean the only thing i
might
add is that as you were mentioning that
is it's almost like
the the predicted input in that funny
sort of way
it's like so we've got what was what's
the predictions
of what the sensory input is
if i could sort of somehow infer that
and that that's that dynamical so i
i think this is really helpful though to
go through it this way yep
and to give a note on that a lot of
times when people think
statistics they think descriptive
statistics
like we're going to take the big data

set the big you know image data set and
we're going to describe it with a
summary statistic
and that does map onto this topology
like observations get mapped through
likelihood onto the most optimal state
of the world
but where generative bayesian modeling
comes into play
is that that model that's inferred that
hidden state what is the real
temperature or what is the real average
height of the classroom
it can also be used to generate and to
combine with other models to generate
testable distinctions you can generate
data
and then this duality where you can like
take a little bit or a ton of data
and then update your prior already using
the bayesian terminology and then use
those prior estimates to generate more
data
that is like expectation maximization or
em
optimization or the parametric empirical
bayes approach
so it's not just taking big data and
crunching it into a regression
coordinate and saying all right we found
the most likely one
it's done this is actually a framework
that allows us to bounce back and forth
between summaries of big data
and the actual large observations
themselves and then here you can think
of the prediction is on a single
observation with a single hidden state

and here it's like
getting a vector through time it's then
you can imagine that it would be
more dimensions but this is just the
skeleton so let's keep on
walking through yep let me interject
here really quick sorry
so can you just go back there we go i
love this what the authors how they
describe
perception i thought it was very clear
they said perception
corresponds to a process of inferring
which hidden states
or posterior expectations
provide the best explanation for the
observed outcomes i thought that that
was very clear
cool and the soft max is the sigma
and that's just saying it's like a
decision-making criterion
uh function but we can also ask the
authors more because that's sort of the
machine learning
uh term yeah so just one just one very
quick thing on that as well i like what
what blues just said there as well is by
bringing it on the outcomes
as opposed to that that outcomes brings
in action
so you you you because the out it's it's
like the perception it's not
what the data is telling me it's also
the outcomes
of what happened in the world with the
body
and that that is really quite i don't
know if that's been so easy to bring in

before

um in the language so i thought that's

quite interesting

yep and we're going to bring policy in

in a second but for now

you're right we have the difference

between the outcomes and the hidden

states

is it raining or not that's the hidden

state and then is my shirt wet or not

that's the

you know sensory state but we have why

is measured wet right

is my shirt wet yeah exactly but we

haven't gotten to this

policy level which is should i go

outside so table one

is just describing in a little bit more

detail

the p and the q distributions

that they use um in their figure one for

this like basic

level and that uh comports their

math with just the broader modeling

bayesian statistics

letters used although the a b c d are

sort of

specific for active so let's go from

that

uh m2 model and

bring in policy so this is m3

model this is the next click that

actually gets us to the first thing that

looks more like a real active inference

model beyond just a bayesian time series

approach so maybe blue go for it again

with what you would say for m3

so for m3 this is the third step and

this is the generative model of
action so here it's inferring in
phenotype congruent policies
from all of the terms that we had talked
about in
 m_1 m_2 and then this brings in policies
which
that's π policies and then e_{π}
is the baseline prior over the policies
and g_{π} is expected free energy
 c is phenotype congruent preferences
and then f_{π} is the variational
free energy
conditioned on a policy so the policies
are
here already here and we just have these
uh m_2 is the what's perceived over
hidden states
uh based on perception m_1 so this is
kind of just adding that
that ex extra layer including policy
into the into the action model yep so
it says here posterior policy beliefs
are informed by the fit
between anticipated and preferred
outcomes
while at the same time minimizing their
ambiguity
so to minimize that so
nice thanks for this a connection
between variables
in this kind of a graph represents them
having like a statistical dependency so
we can look at π
which is policy and then ask okay so we
had m_2
which was this time series that we that
looked kind of like a i don't know like

a

rake or something or a horse and then
policy π gets linked in
not directly with the thumb on the scale
influencing the hidden state
but actually influencing b_{t+1}
which is the transition matrix through
time
of that hidden state so one thing
because there's no π onto the hidden
state you can't have policy at a
snapshot
like timeless policy all policy is
intrinsically related to the movement of
time
because it's hitting the b matrix not
because it's not plugging into
a hidden state directly so you're always
talking about a temporal model just how
deep
are you talking and then it also
conditions the estimate of how the
hidden state evolves that's what that
vertical line like given you know a
given b
so this is we're going to do the time
series how how much
um bitcoin will i have under the policy
of not acquiring any
zero easy estimate easy heuristic
so then those kinds of conditionalized
estimates of different kinds of things
can help agents
act in the world so that's why policy is
interesting to think about and so
instead of thinking about how to
optimize a massively
complicated scheme for maximizing

utility with an unknown variability
this is allowing the agent to model how
one is embedded in hidden states
estimate or changing through time
so that's the π part and then also this
is kind of explore exploit
so minimizing the ambiguity is one
imperative
but also you want it to be fitting the
whole time
like i want my shoe to fit but i also
want to figure out where it doesn't fit
but i need to be walking the whole time
and so by keeping a bunch of these
features
linked in the model the dials
at least exist to modulate in really
creative ways like e is this baseline
prior
and so you could modulate it so that the
prior
is relatively unperturbed by sensory
outcomes
or you could make it very sensitive to
specific kinds of sensory outcomes
so this is again like a modeling
skeleton
that potentially could be used to
evaluate
how different sorts of behavioral
phenomena
like approach and recede or something
like that or freeze
whatever it happens to be might be
linked
by the organism seeking precision
on an estimate of how things are going
to change in the niche given its policy

so of course there's a lot to say
because that's kind of an interesting
area
but that's just the third level so we're
integrating policy decision
into how states are changing through
time and then policy decision
is the interaction of several variables
the baseline prior like the affordances
the habit
as some have said it here and then
c the preferences phenotype congruent is
like

if i'm a shoe size 10 i want a shoe
size 10.

then if you have a phenotypic preference
that's different than what your actual
niche or needs are then it will be
ill-fit so you won't be fit for your
niche

so then the niche will be filled by
those that are fit to it
and then f of pi is similar like it's um
calculating variational free energy
which we can definitely ask the authors
more about

and see how that is related to
calculation on a specific policy you
know

yeah okay any thoughts either of you
well just one thing i think is quite
helpful as well

um is they they call that perceptual
evidence uh the the variational free
energy conditioned on a policy
so it's like um and then that gives you
the sort of perceptual evidence and then
the actual

action model which is kind of like the
expected

free energy and that helps sort of
um show how free energy is coming in at
this kind of

larger time scale um and i suppose one
thing just to mention to the viewers
there is a whole load of nested free
energy processes going into all of this
process but at this scale that we're
talking about

it's so that we can understand in terms
of policies

that we're trying to understand
sounds good and so yeah

here you know um anyone who has small
children knows that their feet grow very
fast right so when we're talking about
the action model

expected free energy it's it's the
expected free energy because i don't
like the actual free energy can't really
be measured in in this situation so
we're dealing with expectation
so there's this phenotypic risk
component on the left side of the
equation

like how much money am i going to spend
on

these shoes and minus the the perceptual
ambiguity like i think

uh my kid is a size 13 but they might be
a one

or two how expensive are these shoes do
i want to buy shoes that fit them right
now or do i want to buy shoes
that they can wear two pairs of socks
with and wear for a little while so

you're always
weighing out this this risk and the
ambiguity that
that occurs okay in that specific
example
the specific policies are like okay i
could ever
measure the feet every day and that's my
sensory outcome the hidden state is the
true
size of the foot and then every day my
policy could be if they're in a new
shoe size we change that day another
policy with a different
trade-off structure different
measurement requirements might be
every two shoe sizes another one might
be when the perceived
ill fitness is too high the squeaky
wheel gets the grease
then we have an action-oriented response
policy
and i'm going to be thinking about all
the outcomes
in this model and about the certainty on
it so
nice yeah i mean it also relates to that
when my daughter has a big toe starts to
get sore
that's my perceptual evidence it's like
okay it's like
that shoe is rubbing a bit now so yeah
at some point i've got to make a choice
so that's a good that's a good example
actually
it's an error signal that it's ill fit
into the policy and that's a niche
modification with a shoe

um but it's a similar idea so table
three kind of goes through
m3 in a little bit more detail
and that brings us to figure three which
is the fourth
click of the active inference uh
kind of chain here and it adds just
two more letters it's mostly the same as
figure two
like the c g and the e which connect at
pi
that was my mistake in the 18 2 and i
connected the e to the um
g instead of the e to the pi um they're
connected
at the pi as shown here um the c g and e
are now in m3 with a blue
box and then m4 one more dependency is
being clicked in here
which is gamma which is being clicked to
g
so gamma is now an input to like a
bigger g
function so the each the g is
is a function um which you know you
could input just
a number you could input a few depending
on how you write it so
this is a slightly more expressive
g function because it includes this
precision
variable and gamma and b are kind of
like one over each other
the reason why it's kind of like when
people take logs or do one
one over or negative this it's just
sometimes you want to get the most of
something but it's easier to frame it as

getting the least of something
or vice versa um but what this gamma
represents is the expected precision
because um this is kind of certainty so
this one is uncertainty
so what would either you say about this
final

click to m4 blue or steve so this is so
this is so fun
so the gamma is like how much did that
shoe not fit

and the beta is like how fast are their
feet growing right

so i think that that's the i mean i
guess beta doesn't take into
consideration time but it was
um it was described as a rate um
it was described as a rate so maybe it
does include time i'm not sure
i don't want to go too too far down the
rabbit hole with the shoes but it's
almost like

there's more precise shoes like you'd be
more precise with the steel toe and then
you could have the more stretchy
no no no no one take this too seriously
it's about what it actually does
say which is basically the um precision
which is the expected precision is in
the gamma

so that the uncertainty can be yeah in
the b

and this figure three the reason why
they even went to this level of
incrementing it out first off is to
show us level by level how we're
building on

okay it's about a time series that's

being predicted through time of sensory
outcomes
that are like generated from an
underlying time series of hidden states
vis-a-vis a likelihood mapping and a
transition matrix
and that's getting played into by priors
at that s1
level at the first state level and then
there's policy that's the whole
cybernetics control theory and then this
is the machinery of the decision making
and now we're just adding
one more level which is the implicit
metacognition
so now we're adding this implicit kind
of always on
always being integrated into the model
again not a
disaster mode recovery mode for the
otherwise
non-affective agent but it's an
affective agent
whose implicit metacognition its
kind of key regulator here is related to
its
certitude not the thermometer's tracking
the average
of um it's its returns on investment or
its average estimate
but its uncertainty in its generative
model
so that's what we get to with
their figure three with a fourth click
so wait wait i think we need to talk
about here uh because it's the first
time that it comes up
is there um affective charge

here oh yes so in this in this model
and which also m_4 really is the top
level illustrating like the hierarchical
um hierarchical nature or the nested
nature of how this model is
is normally and then in the effective
charge

it's it's described here as phenotypic
phenotypic progress which is the policy
in the prior minus the policy in the
posterior

and then the free the free energy of the
prior posterior minus the prior wait no
i said that wrong

prior minus the posterior there it is
yep here here it is in table four

so let's look at what they say so
expected precision is γ

okay looks like a little dart γ
precision

you want more precision in an optimistic
world model

and policy π okay and it turns out that
 γ the expected precision is one over
a γ distribution that's gonna be the
beta

and that's just gonna be a statistical
distribution that

plays nicely with statistics just like
you see like natural log like the γ
the normal these are

tools that statisticians use and so they
kind of

work here but we can ask the authors to
what extent does it have to be this way
but anyways γ is the expected
precision so

γ and β we can kind of think

about them as like linked because again
they're one over each other
so wherever you see a gamma or beta it
could just be replaced with the other
one
and affective charge is being um
defined as yeah the the uh change in the
policy
multiplied by the free energy of that
policy
so not just if it's good but
unchanging
then this is gonna cancel out so the
phenotypic progress is low
if you expect you know the 10 bitcoin
and you see them
and you're not changing your policy in
this
formulation that's why we don't need to
tie it too closely to a specific
emotional experiences because those are
going to always be super
subjective and require a ton more
specification but we're just kind of
broadly calling this ac variable
something that's related to both the way
that the phenotype is changing the
policy is changing
and the way that the free energy
estimating of that policy is changing
but there's something important to note
in that equation is if the policy
doesn't change
then the affect of charge is zero
yes and let's again let's have the
authors give us
the authoritative um uh information on
how to read some of these terms

but here is what they said about that
step four

the fourth level which is again this
gamma precision

is required to enable the system to
estimate its own success which could be
thought of as a minimal form of implicit
non-reportable metacognition so even to
be able to

semantically express metacognition like
i am unsure that sentence

is many levels or another level
away from just this precision variable
which is like an in it doesn't have a
self report

circuit this is just like a model that
has an error term that it's
updating at a slightly higher level than
just the likelihood mapping and
the expected precision term that's gamma
reflects

prior confidence in the expected free
energy over policies
the expected precision term modulates
the influence of expected free energy
and policy

selection relative to the fixed form
policy prior

e sub π which is the habits or that the
prior

so it's kind of like should i fall back
to priors

or should i go with what

the g function lean into what my
 g is telling me then

um yeah formulated this way we can think
of

of gamma the precision as an internal

estimate of model fitness subjective
fitness

so higher precision my internal model is
doing better

lower precision my internal model is
doing relatively worse

than it could be it represents an
estimate of confidence

in m4 that fourth click in the phenotype
congruent model of actions

given the inferred hidden states of the
environment so they kind of

tessellate it back out there they're
saying

it's about those outcomes and about what
generates them

and how those change and how our
policies play into it and

it's about how certain we are in that
generative model

and there's a lot of ways the
environment can change so if we're
getting good data from our observation
then we should be reasonably confident
with staying with what works to some
extent

but if that should change there should
be another mode

that allows us to rebalance
preferences affordances um

and integrate that in a different way in
a different allocation

to our policies

and then to emphasize its relation to
valence in our formulation remember

valence is like feeling good or bad

positive negative going forward we refer
to gamma

updates using this term affective charge
so here's α affective charges change
in β aka change and γ in a just
another way of saying it it's change in
precision

and then you can imagine if if the
precision goes up it's going to be good
precision going down bad and flip it for
the one over or however it ends up being
written

but that's what it's getting at when the
precision is going up

as new data are coming in like they're
you know i'm thinking of a slot machine
and then it's like oh you won something
and now the second digit is going to
spin

so there's both an increase in charge
and then that like kind of restabilizes
and now

it's about whether that changes again or
whether the second digit is better
and so that is how affective charge is
being tracked through time
with these γ updates

okay any thoughts or continue 32 well i
think well i think the one thing that
sort of i find helpful here in this here
is

it's it's the general overarching active
inference

um parameter of uncertainty

is like it's it's they're using the term
confidence and precision more at these
higher levels because it's like a
meta intern is still at some level
uh an interpretation of uncertainty but
because it's on policies in that

confidence and precision
are at play because you're interpreting
your ability
to act not just the raw
sensory sort of mellow so to speak
right there's a big difference between
like visual uncertainty it's just
two shades of grey and they're so
similar you just don't know what you
don't even see a difference
versus two strategies two chess moves
that are very different and you're just
not sure which one to do
that's a very different situation for
inference
because the way that your eye scans on
the page
is kind of reversible you get you're
getting to retract your
your choices but with strategy it might
be very irreversible very non-linear
um okay here's another example they give
on 32.
so this carrying on right below equation
3.1 which was
screenshot there so um
ac can only be non-zero when the
inferred policies are different from the
expected policies
so blue that's what you had said so if
it's sort of
at the bottom of the bowl from a policy
perspective thermodynamically
it's not moving the delta is zero it's
positive when perceptual
agents when it is positive when
perceptual evidence favors an agent's
action model and negative otherwise

so it's like are the data coming in
locally
good for my model or locally not
congruent with my model
in other words positive and negative ac
corresponds respectively
to increase and decrease confidence in
one's action model
so as the data are increasingly good you
actually like lean into your model
and prioritize that prediction and then
you kind of
start maybe overfitting and then maybe
the model has to pull
back and rely a little bit on something
different
to to re-equilibrate that to stay on
this multi-dimensional front with model
fit
and optimal learning so that's
one of the ways that active inference is
rethinking the explore exploit
is by relying on the generative model
and then the trade-off with different
ways the generative model can get
combined with priors you can get
explore and exploit behavior as well as
a lot of other modes and a rapid
switching that's informed
by the instantaneous performance of a
model
um then they give some examples of
different a prey and a predator animal
okay and then here's three any yeah any
thoughts on that before five three
okay five three affective charge lies in
the mind of the beholder
so that's cool because it shows that the

authors in this research
line have taken up the
challenge of ecological psychology or
of action-oriented philosophy or
whatever it happens to be it's like
this relational insight that these
variables have to be about the beholder
and their relationship to a situation
partially for computational
tractability and partially just because
it's kind of
how these systems seem to be we are
modeling things that are in the mind of
the beholder so it's always going to be
with the situation and the context and
we're always going to have to go in
to the specifics um
but we're also going to find the
patterns that are similar for example we
all
as they say tend to experience a rush of
satisfaction
not five hours of satisfaction unless it
was a long jigsaw
when we solve a puzzle or understand the
punchline of a joke or act of inference
an aha moment our explanation is
straightforward
in active inference biologically
plausible forms of cognition
inevitably involve policy selection
policy as fundamental so policy is not
the disaster mode this is as fundamental
as affect
whether those policy internal selections
the selections are internal
citation citation citation or external
period therefore ac is also elicited by

mental action

typically in form of top down modulation

of lower level priors

i'm getting the lower level pain signal

i shan't not think of it

and then now i'm feeling better

as that is top down prior is influencing

the bottom-up signals just to give one

example

so this is like bringing a lot of pretty

interesting areas together

especially because again it's not just

about humans as cognitive agents this is

about

a scale-free formulation for how

affective charge works in active

inference

but we can look at the specific example

that they do

provide with the simulation

so blue

any thoughts here or on figure five as

far as what they do in the simulation

um maybe should we go through the

simulation first do you have that figure

up or no

yes i think it's is it this one

yeah figure four okay sorry yeah

yeah i think that yeah that's probably

better um

so in this simulation the way that they

set up the teammates

is um it's a rat or it could be i mean

it's a simulated agent right so it could

be a router it could be an ant or it

could be

anything in a maze where on one side of

the maze is food

or so the mouse is placed in the middle
of the maze and
on the north end say you have on the
left side food and on the right side a
shock
and on the south end there is an
informational cue
about whether the or about which side
the food is on as opposed to the shock
and so the food location changes over
the trials uh and then the um
the subject is allowed to move
themselves and so
typically the subject checks the uh
queue and then gets the reward so
there's like some
there's a uh and a reward
for like you don't get shocked
accidentally if you first go see which
side
that the food is on so how the cue is
associated
how the cue is associated with decision
making well if
if they know to epistemically forage
like that that's more valuable than
you know accidentally getting shocked
then they go to do that but i think in
the trials they changed they kept the
location of food consistent
for many trials and then they changed it
for for many trials
and so that the mouse had to relearn
what to do
yep so let's just imagine the food was
always on the left side
never changed well as soon as you went
to the left side 10 times you'd be like

i don't even need that q anymore it's
kind of useless info gain it's going to
be not good
time spent so i should just go for the
policy of going
right for the food now if it was
switching every time
it would make sense every time if the
shock was painful to get that cue
every time so the mouse is able to
in one way make a binary decision which
arm to follow
for the food but also there's this
higher order decision
about getting information and actually
in the active
infra infrarents the ant paper that i
did with some of these
um co-authors we also used a tea maze
but there was no
metacognition there was just the
stigmargy of the colony
so the teammates is really a
generalizable framework
and once you have the code to make the
teammates
you can just make basically anything
it's just like
any other shape maze is just another
matrix so this is again
it's like an initial foray and the
places where people see that it's
lacking like that's where we want to
kind of
build in the gaps let's get to a little
more
formal view of how you make
that active infrarat

that is doing the act of inference on
this foraging
decision so this is a there's a bunch of
sections
to this figure and
blue would you say it's better yeah i
think five is better before seven
right yeah okay so there is um
a generative model uh
first off i mean there's a lot of entry
points people should check out the model
and just read it themselves because it
will take
there's a lot of ways to enter into it
but one thing to notice
is this c the preference over outcomes
if it's not if it's food it has a plus
four points shock is negative six
these are just numbers you can make it
scaled differently and then if it's not
a shock or food it's
zero and then also it doesn't have a
preference for going left or right
so these you can already see are some of
like the little atoms
that are going to get fed into a bigger
model
so this bigger model is going to be able
to take
kind of these variables like how much is
the relative weighting of
the food versus the um lightning bolt or
is there an implicit preference for one
side versus another and then here the
prior d
the priors that they're 50 50 that it's
on either side
but again that's actually different than

a preference
for left or right branch is different
than a belief
that it's likely to be on the left or
right branch
so we can separate out those two
variables i wish it were this
temperature that's my preference
but i'm very accurate that it's not that
and that's why i'm sad
so then you can talk about those kinds
of combinations
of different parts of the model
and uh b so that's d and c b
is the state transitions and so these
are
it's kind of like a second level matrix
and this is kind of it's it's like a
cube
if you think about it's like a four by
four by four because there's four
positions it can be in
the waiting position that's this front
one
then it can go to left to the right or
to the bottom
and then there are state transitions
that are defined
like if you go from the center to the to
the left branch
that's going to be a defined move in a
matrix that you can make
then a is the mapping between where
you're at
and whether you get the cue left or
right
so that's the info that you'd get on the
bottom the south branch

and then the reward or the punishment
and that's the likelihood mapping
whether you go to the left or the right
regardless of whether you got the q or
not and then e is that baseline prior
over policies

which um is just it's even here they're
all 2.3

we could ask what is the effect of
different values it might make
you know you could work that through in
your head a little bit

but it could be different but here
all of the expressivity is in the other
parts like the preference for the um
reward

over the shock so one doesn't need to
specify like

yes and i'd rather you know take a left
turn at this exit while i'm going there
but that could be specified because
again it's just sort of a

um skeletal model of this scenario
any thoughts on this before we go to
seven

nice but isn't it it's cool model and
it's nice how even with only four
locations there's so much richness in
the behavior

so shows you don't need the little
scurrying animal

to simulate the spatial dynamics of the
maze because then you're going to get
trapped up

oh is it a narrow chamber or a broad one
but this helps you

look more semantically perhaps at the
way that the agents interact with their

environment

okay so now on seven

um blue do you want to say the two halves of seven

yeah so this uh part of seven this is the

the generative model that they've created for aspect of inference and the next slide is going to talk about all of the different

uh matrices that go into producing this model so that's why i don't know maybe it's this one maybe this is better first

go forward um so the uh here the the different matrices that go into the the larger model in figure seven include the prior expectations d^2 for any initial states so that's this uh top left and then the second one which is down here on the bottom is the affect of prior which is the likelihood matrix reflecting some degree of uncertainty in the predictions which was

also shown in figure 5 multiplied by that beta

and sets the expectation on expected sets the expected precision so which varies between

negative point five which would be a negative um

a valence and then uh two which is positive valence

uh and then this contextual prior is this matrix

aac a to the c

for the likelihood mapping from the context states to the lower level uh reflecting the fact that the agent is always certain which context they observed after the trial is over so they either got the food or the shock and they know that right and then this last one is the state transitions

b b2 b to the two um and this is state transitions at the second level reflecting a couple of assumptions that both the affective and contextual straight states vary but have some stability across trials 0.2 to 0.3 probability of change and that the agent has a positivity bias in the sense that they are more likely to switch from a negative to a positive state than vice versa

thank you blue so what's happening in seven is we're adding um another level so here we see that beta and gamma being connected to the whole cge policy and then that rake formation and now we're going to tie in this implicit metacognitive layer with another hidden state so that introduces this parentheses notation where if there's a parenthesis it means which level of the nested markov decision process it is so that the s that we knew and loved that was giving rise to the sensory

outcomes is now s_1 one
and that's the first level of the model
and then
the s_1 that is given rise
to the precision is
 s_2 the hidden state so that is now
what is being called the implicit
metacognitive component
which is the second level of the model s_2
upper score two which has its own prior
 d_2
upper two just like this has a d_1 upper
one
it's actually not shown here but it is
there um
there are uh the same
markov decision process structure is
being recapitulated
but there's a different interpretation
because the hidden states at one level
are now the observations linked through
an a at the next level
so here's the observations like what
square am i on
the sensory outcome am i and then a
hidden state what kind of square am i
on and then that's getting mapped to
am i happy or what that's the whole
question of this paper is what does this
actual
second state yep stephen oh yep
now yeah i'm gonna i'm gonna have to
head out sorry thank you peace thanks a
lot yeah thank you bye
so then you get all these kind of
interesting uh
different kinds of matrices at the
second level like

given that i went left and the food was
left

then will i find out about that and so
you need a few matrices to kind of close
the doors and make the model run
but that's a little bit on the
implementation details and we'll
probably learn

from the authors and in 10 we kind of
just have a summary
this is i think a great figure because
it shows how multiple time scales of
nested processes are linked
from the inheritance of prior phenotypes
like how big

is that foot and also development
learning like how fast does it change
get linked in through multiple levels
that are getting nested closer and
closer to action and perception
in the loop and then rippling back out
so pretty nice overarching
figure any thoughts on this or continue
on blue

this just really like like like you said
it's a nice figure a good summary and
just reflects like the deep
temporal like nature of this model that
they've

built next let's uh go through the
results of the simulation and then just
deal with a few
little further implications and then
that will be a nice

19. so in figure 6 they
simulate out that scenario that we just
talked through a few times with the maze
for 32 trials

with food located on the left side so
this was just testing
that it will converge to expecting the
left side
this was kind of like a little sanity
check for their model
but also it shows that we're doing this
in the in the broader spm toolkit
and so that means that a lot of these
things
like the um ability to visualize and
easily run these models
and check that they're complete and
debug them and run them at scale
a lot of these things that tie up a lot
of innovating
innovative modeling efforts a lot of
those are simplified somewhat and
again we'll ask the authors about this
um
so they simulated in figure 6 just to
show that they could do different
policies from the exploratory policy
in the first 12 and then switching over
to just the go left
policy okay figure eight
anything to say blue
no okay yep so figure eight they're kind
of just
giving a few trajectories of what can
happen
so it's it's characterizing the model in
a little bit of a playful way but i
think it's
fair and as we'll see it's it's at least
computationally warranted
so here we have initially anxious
the state is negative for the first

eight time steps

whereas getting the q and the left side
and then it develops confidence
and then the green line here in eight is
where

the um the situation changes
so then there's a few time steps where
it still hasn't

caught on before it gets shocked and
then

it gets anxious again and it gets
renewed confidence

so this is capturing a few of the
just like with the ants how we showed
how it could switch arms

here they're showing not just that it
that that it can switch arms from
q and left left and stuck then the shock
unsticks it it's like whoa i was very
sure

that going left was the right decision i
got shocked

now i need to go take the q and not just
one time i'm going to get the cue
multiple times

okay i think that i have renewed
confidence in this direction

so it's really nice model behavior

yeah what do you think blue so they did
um

actually there's one thing and so it's
not showing here maybe it's gonna show
in figure nine but maybe they

um maybe they i don't know if they
showed it in the figure or maybe

we'll just let the authors talk about it
yeah

so about the model that lacked the the

last level the metacognition about like

so

in one of the models they took that that

highest level of integration away

um to see if the mouse would similarly

adapt so here they changed the position

of the food right in this in the top

right figure

so they changed the position of food it

was always on the left always on the

left always on the left the mouse

adapted it didn't even go look for the

cue

and then in the the model without that

metacognition the model thought the

metacognition didn't adapt to going to

look for the queue like they just like

stayed stuck in the middle like not sure

do i

i don't know if i should go up or go

down i thought that that was really like

funny so so this is what happened with

the metacognition initially anxious

developed confidence

anxious again and then renewed

confidence but in the model that left

that

uh that integration that metacognition

it didn't it just like stayed stuck in

the middle i thought that that was a

very like

cool anomalous effect and like a really

neat control like

trial for for this um test nice

and it actually relates again to this

idea that affect or anxiety is

in the loop and it's something that's a

continuous variable

and in uh you know healthy settings we could say maybe anxiety is actually warranted here there was a shock in a strategy that was pre there's nothing wrong with a strategy that works and there's nothing wrong with switching strategies when it doesn't work but there is some variable whether we associate it with suffering again that's a second question but there is some feeling or some computational need for the system to actually pull back and re-weight how it uses its prior evidence versus how it deals with incoming evidence so i think this is a really nice example of just the categorical the discrete kinds of behavior that are very different from low precision and negative valence and then a kind of phase transition rapidly into a stable phase into negative and back again and integrating it with decision making with as they say early on the imperative is that reduction of uncertainty with the generative model not maximize reward not minimize suffering either but actually to optimize its precision aka reduce its uncertainty about a generative model of the states of the world including its own action so it's just really

nice work some of the reasons why
active inference is so cool actually and
then figure 9
they simulate 64 trials oh i think this
is what you were speaking to
an affective agent plotted in orange and
an agent without higher level contextual
and affective states plotted in gray
so there's it's um i mean one could look
at figure 9 a little more closely to see
the difference
but yes the expectations in like just
looking at the top one
the expected food location they both
converge to left
and the per the orange goes a little bit
higher so it learns a little faster it's
like yeah i'm doing really well on this
test
this is an easy course then the green
line happens
and the gray one slowly updates it's
like hey i've been right for you know 20
and i was only wrong twice
so i'm at 90 percent what's the issue
and the orange one's like whoa
something's wrong
because i'm getting error signals when i
wasn't so and the green line here
represents the food changing right like
that's the context switch i just wanted
to contextualize yes
yes the context reversal so it was on
the left and now it's on the right aka i
got a shock
and so then also look at this the
affective charge the orange one
anxiety drop

uh blissful ignorance anxiety drop and
then the gray one
doesn't have affect so we don't see it
there but then
um yeah or the gray ones uh affective
charge is actually a little bit all over
the place but we can ask them what that
uh means and uh so yeah
pretty it's it is super interesting
because like learning you know
even organic chemistry is an emotional
learning process
so how actually affect and policy and
mental effort and all these things come
together it's kind of cool
i wonder if this is like just like to
take it to the like
meta level like is it like um
you know self-reflection right like what
at what level of integration is that
right like so you do the action you
select the policy and i
think that my doing team mazes don't
self-reflect but like
when you integrate at that level there's
agents that do humans that do
self-reflection and then humans that
just kind of don't think about it right
and so what what is the better
adaptation there it
just it provides evidence for that i
think yep and similarly it's like
you know i want to be doing the right
thing
but then will that preclude you from
having a thought that where
that's actually being um that
expectation is being violated especially

if there's sensory evidence that that's
the case

those are really interesting situations
right so like psychopathology like are
they just missing

some higher level integration like for
to

lacking that affect you know

well ask the authors nice so then

um just the additional info section of
the paper as we kind of close out

in the last little few slides here there
are four appendices

and the code is available on casper
hesp's github

and so you can see just from the top
lines of the main

matlab script yeah it's in matlab it

uses markov decision processes

and it involves active inference and

learning using variational base the
factorized version

in the spm toolkit using matlab so check
out the model streams and

let us know how it goes to work with the
code

we kind of just had one or two more
random

areas to cover a few directions that
this takes us in

which we've discussed a little bit one
of them is more advanced

simulations so can we have two

dimensions of affect or could we have

another summary variable that

represents some other feature of systems
could we

bring it more into alignment with

computational psychiatry and
translational research and could we
embody robots that implement these kinds
of
algorithms how would that change some
type of
situational like physical robot or even
a software robot if it had these kinds
of
estimators internally and then we're
always just thinking about
where does active inference come into
play which is easy for this paper
because that's the whole model but um
where
where are the next empirical steps which
they mentioned in 5.7 a little bit
and how does this relate to this realism
versus instrumentalism question we're
always getting at like
is this what the systems are or is this
how we're gonna model the systems or are
those one in the same
thing um
any broad ideas and questions before
this little freudian
interlude yeah okay so thanks dave for
pointing out um
a lot of the recent work of mark solms
specifically this new project for a
scientific psychology
general scheme which the link's given
down below but it's um
pretty interesting work that has to do
with
integrating recent developments in
neuroscience with some
freudian and uh ideas and beyond

and i guess i just like ideas that
aren't named after people because then
it gets pretty much like oh who's doing
darwinian evolution
what does it really mean to be freudian
it gets a little bit
historiographical instead of about the
ideas
but the mappings that are made
are clear and stand on their own and and
put the sort of history
hopefully um to the side
when we see that there are commonalities
across very disparate models
and here we see a familiar structure
which is kind of like that markov
blanket
separating the internal and the external
states but
the letters are a little bit different
and so there's an abbreviations table
given in
this paper and also
for many years fristen and carhara
harris
and others have been writing papers
that are about different freudian
concepts
and relating it to free energy and drive
and
a lot of other points of contact with
different psychological threads it's
just not uh
an area i'm super familiar with with the
literature and the debates around but
it's an area that's obviously important
to many and influences a lot of areas
and it's just interesting to see how

these developments play into the
absolutely still open for evolution
field of psychology psychiatry
so that was uh pretty much what we
wanted to
cover in 19 so that was pretty fun and a
little little sprint to get
the paper finished because it was
somewhat of a long paper but
it's really worth it so blue any final
thoughts or what
what are you taking moving forward or
what do you hope that we
discuss in the 0.1.2
i don't know i'm looking to really
discussing uh the bayesian brain
integration or or like correlation here
i really think that the
you know ability to update your
beliefs based on what your previous um
experiences is like this is like the
perfect
like case study for this kind of
philosophy and i know like the bayesian
brain catches some slack sometimes and
bayesian statistics catch a lot of slack
but in reality like we
aren't going to go to the shock side of
the box if we know that the food is
always in
the left side of the box like so having
that that update so i'm looking forward
to just see what
what people have to say about the
bayesian brain aspect of this paper if
anything
we get that far if we can make it all
the way through the model right

nice yeah i agree i was also curious
about okay well what if we could just
associate
a like you know like click share
whatever
on social media and then maybe you don't
even need the shock
or maybe you could add that or something
but with that matrix
could there be upvote downvote systems
where maybe the data already exists
and then the positions that somebody
could be in would be like
i have not seen it or i have seen it
something like that they're moving
around
pages seeing or not seeing things and
that's influencing
their beliefs about the world and their
beliefs about themselves and
how how accurate am i if this is what
the info i'm getting
is and then that's
without even going into the
instrumentalism realism at least it's
something that
it feels like to be searching for
information or a question that can be
useful
so it's pretty cool stuff thanks a lot
blue and thanks also stephen for joining
for 19.0
we'll uh see everyone for 19.1 and two
thank you